

Qatar-1b

R-2446

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110531 · created 2026-08-22T23:38:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455712.9534 +/- 0.00045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.04
Transit depth (Rp/Rs)^2	2.47 +/- 1.27 [%]
Orbital Inclination (inc)	84.72 +/- 1.86
Airmass coefficient 1 (a1)	0.99 +/- 0.017
Airmass coefficient 2 (a2)	0.01 +/- 0.42
Scatter in the residuals of the lightcurve fit is	1.67 %
Best Comparison Star	#1 - [554, 339]
Optimal Aperture	2.86
Optimal Annulus	10.95
Transit Duration (day)	0.0637 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.04 vs 0.1463 (deviation 0.22σ)
Duration fit vs published	0.0637 d vs 0.0692 d (deviation 0.64σ)
Mid-time O-C	-1.1 min
Depth SNR	3.2
Residual scatter	1.67 %

Light curve

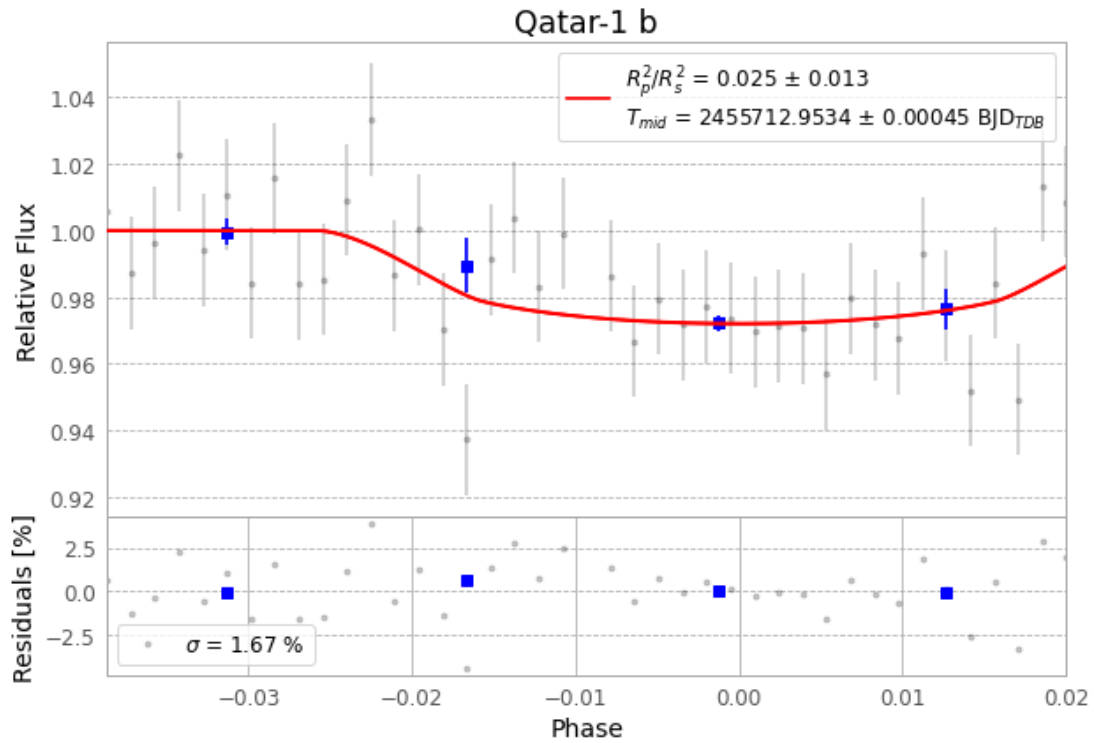


Figure 1 — FinalLightCurve_Qatar-1 b_2011-05-31.png (SHA-256 8ac2a7d855fce630...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [441, 258], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 742c735d8eb0f275...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

50 FITS frames (33 MB), Qatar-1110531093313.FITS → Qatar-1110531120014.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110531.log (c46919da117f...), FinalLightCurve_Qatar-1 b_2011-05-31.png (8ac2a7d855fc...), FinalLightCurve_Qatar-1 b_2011-05-31.csv (8aa927a97d84...)

Compute resources

Wall time 189.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 23:38Z.